Project Guidelines

You may use any programming language or tool(s) to do the course project. Please submit the following documents:

1. A PDF report summarizing your approach, results and analyses
2. Link to a githib repository containing your code and README file

Group size: 2 members maximum (each member's contribution will be evaluated separately).

Please start thinking about a project topic as you will be asked to submit a 1-page project proposal very soon in the following format:

- Project title
- Research question
- Methods
- Dataset(s)
- Key research paper references (atleast 2)

Here are some strategies for executing a good course project. Please do not hesitate to write to me about finding a good project topic.

Strategies for doing a project

1. *Come up with a topic of your own*

2. *Select a topic from the list of topics provided*

3. *Extend any of the assignments you did in the course.*

4. *Replicate a research paper of your choice (see last section of this document)*

Submission Policy (GitHub)

To encourage consistent progress and discourage last-minute submissions, the following policy applies:

¬ʌ Minimum Progress Requirement: You must push at least two meaningful commits on different days prior to the deadline.

¬ʌ Commit Timing Check: If over 80% of commits are made within the last 24 hours before the deadline, a 10% penalty will be applied.

¬ Commit Quality: Commits must reflect actual progress. Non-informative or placeholder commits will not count.
¬ Final Submission: Your final marks are based on the latest commit before the deadline, but commit history will be reviewed.

Replication of Research Papers

An interesting blog by Prof Jason Eisner (NLP+CL researcher and ACL fellow affiliated to John Hopkins University) titled "How to Find Research Problems" classifies CS research problems into:

Big bones *and* ***small bones****(analogy from biology)*.

He recommends **replication** as a pedagogic strategy (with dividends in terms of publications if done properly).

https://www.cs.jhu.edu/~jason/advice/how-to-find-research-problems.html

Time permitting please skim through the postscript of this message containing a link to an interesting page titled Eisner cites: "How to Write a Publishable Paper as a Class Project": https://gking.harvard.edu/papers

(Pasted verbatim from Eisner's blog above)

The biological anthropologist Loren Eiseley used to say there were two kinds of scientists: ***big-bone hunters and small-bone hunters****. (He himself was a small-bone hunter, he said, fitting little bits of data into the skeleton. If Eiseley had been a programmer, he would have called this "bottom-up science.")*

Computer science includes many different kinds of research efforts, some of which are more tyrannosaurical than others. You can contribute to one of these efforts in various ways.

About the smallest bone that you can find in Computer Science is a reproduction or implementation of someone else's work.

A significant small bone to look for is a tweak that improves a well-known technique. (In many subfields, you will be expected to demonstrate objectively that your method is an improvement.) Much research is of this kind.

Along the same lines, you might make a controlled comparison of two or more algorithms,

evaluating them by some objective measure of efficiency or accuracy. Designing a clean comparison does take thought, and carrying it out is often a lot of work.

*This is usually a **medium-sized bone**, depending on how much work it takes and (more important) how surprising the establishment finds your results. Note that quantitative studies of this sort are becoming increasingly important in some areas of CIS (e.g., operating systems, machine learning, natural language, algorithms).*

*Your field identifies various problems or issues as significant. These often represent **big bones** in the skeleton of the field -- problems that arise often, and whose solution makes a difference. Get to know some of these problems and the work that's been done on them. If you see how to achieve the first-ever solution, or a better solution, or a different style of solution, that's a big deal. Sometimes finding a good solution involves changing the problem slightly.*

Eisner's blog above cites the following references:

King (2006) describes "how to write a publishable paper by *beginning with* the replication of a published article. How to Write a Publishable Paper as a Class Project: https://gking.harvard.edu/papers

This web site shows how to write a publishable article by beginning with the replication of a previously published article. Following the advice here constitutes a central assignment for my class, Quantitative Social Science Methods, I. After fine tuning these suggestions over many years, I published the 2006 version as:

*Gary King **Publication, Publication,** PS: Political Science and Politics, Vol. XXXIX, No. 1 (January, 2006), 119-125 (Abstract: HTML | Article: PDF)*